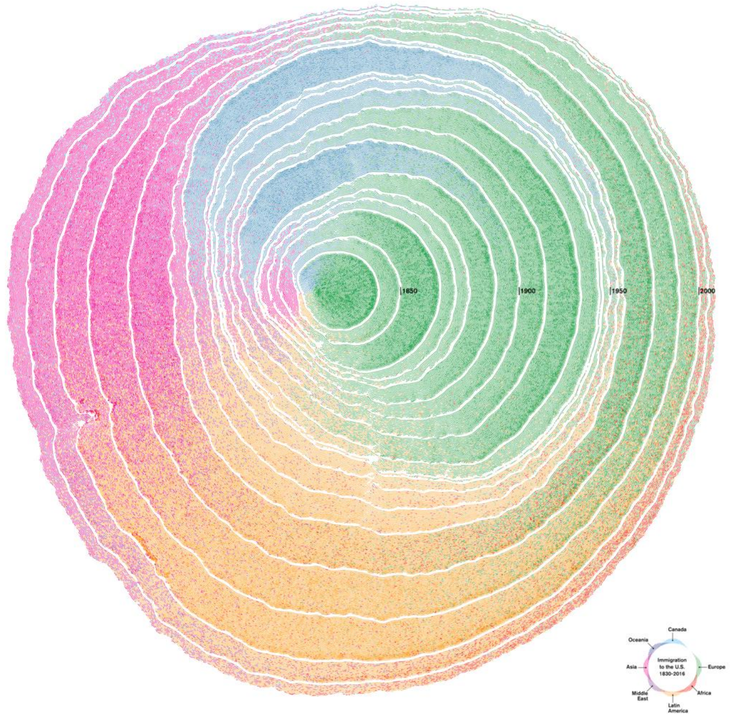
HSLU Lucerne University of Applied Sciences and Arts

Data Visualization for ML and AI

Dr. Teresa Kubacka



National Geographic, Northeastern University

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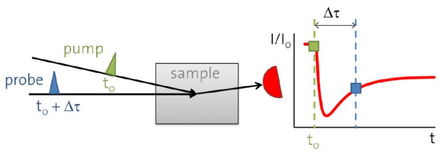
20. Februar 2026

FH Zentralschweiz

Welcome!

About me: Dr. Teresa Kubacka



PhD in physics, ETHZ

data scientist
since 2016

Currently working at the ETH Zurich (analytics for research mgmt) and Xurce AG (freelance)

My focus:
- building data-science-powered tools and services
- advanced analytics
- data visualization and storytelling

https://www.linkedin.com/in/tkubacka/

https://github.com/paniterka/awesome-matplotlib/

https://www.teresa-kubacka.com/

Who else is involved?

Philipp Meschenmoser

HSLU

full-stack developer,
data scientist in Smart Region Lab

teaching assistant, technical support

Barnaby Skinner

Neue Zürcher Zeitung

Deputy Editor in Chief,
data journalist,
head of NZZ Visuals 2020-2024

guest lecture on data journalism

Tell me about yourself!

- why do you do Bachelor in AI, which specialization have you chosen
- your expectations of this course, your hopes and fears
- if you wish: your best and worst experience with data visualization

What you will learn in this course

- You know how human perception influences how we communicate with data
- You know the main principles of turning data into visuals
- You can critically evaluate and analyze visualizations
- You know and can apply guidelines for designing a good data visualization
- You know how to pick the right tool for a given data visualization project
- You can independently design and develop static and interactive visualizations

- The official language of the class is English

Focus

- Best practices and transferable skills
- Relatively wide overview
- Focus on data visualization for data science
- Many examples and inspirations
- In-class and async work and discovery
- Coding using tools in Python ecosystem (Matplotlib, Plotly, Streamlit)
- Minimal set of knowledge for you to start building upon

- Every now and then, I will give you links to extra material if you want to dig deeper

- **Have fun!**

What is NOT part of this course

- Dashboarding with tools like Tableau, PowerBI ...
- Research papers
- Web development
- Customized interactive visualizations
- Javascript
- Databases, analytics engineering
- Statistics
- Data art

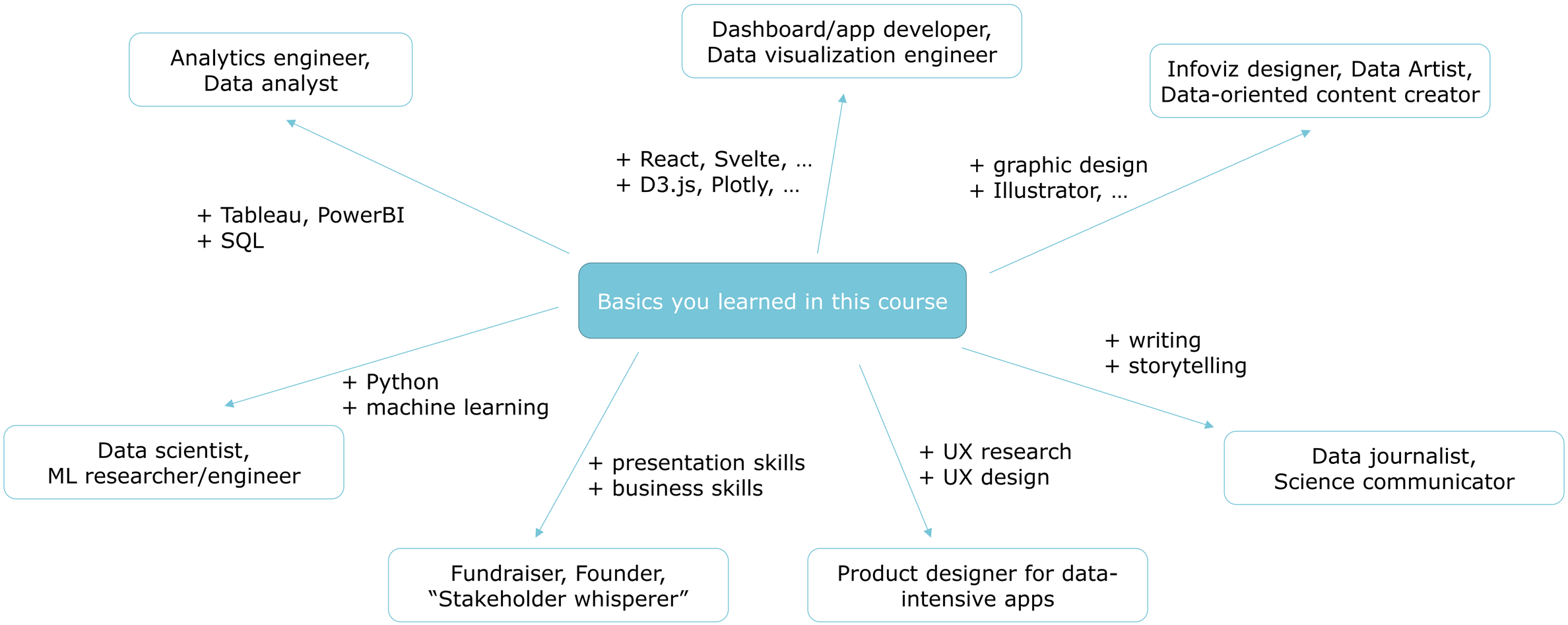
Dataviz is interdisciplinary

- You will notice that the course content overlaps with many fields:
  - Analytical thinking
  - Coding, data wrangling
  - Graphical design
  - UX research, UX design
  - Storytelling
  - ...

- You will notice that there are more guidelines than laws of nature: it is not an exact science!

- Your feedback about how useful you found the content will be very welcome (it has no influence on the grade ☺ and can be anonymous if you wish so)

Some examples on how you can build up on this course:



What and how you will learn – more in detail

- Theory and exercises
  - Visualization as communication
  - Visual encodings, human cognition, chart types
  - Intro to Matplotlib ecosystem, special chart types
  - Styling, annotations, elements of graphical design
  - Colors, strategies for dealing with complexity
  - Interactive visualizations, elements of UX for dataviz, intro to Plotly
  - Other Python tools, introduction to data visualization on the web
- Including 2 extra topics:
  - dimensionality reduction
  - data journalism
- 3 weeks of individual coaching (20-min sessions) and project work: this is where the real learning happens
- Preliminary project presentations with peer feedback session

Semester schedule

Mini-project:
- due by Tuesday 31.03.2026

Final project:
- define teams and topics by 17.04.2026
- interim presentations on the 29.05.2026
- final submission by Sunday 14.06.2026 (just before the exam session starts)

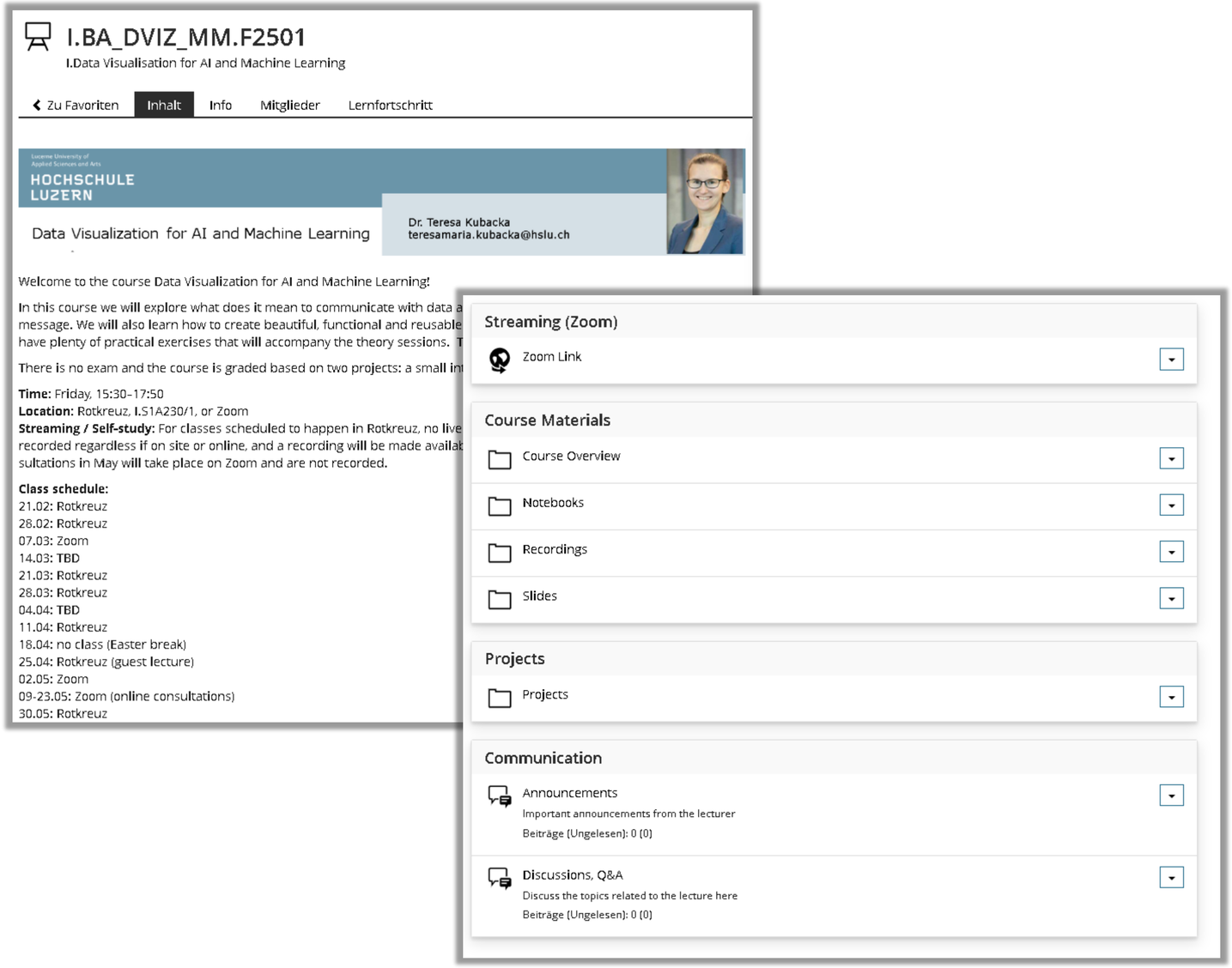
Learning Materials

- Ilias

- Individual feedback: per email, teresamaria.kubacka@hslu.ch
- If you think other students may have the same question, don't write me an email but post on the Forum, so that others will see my answer as well
- Each lecture, last 15-20 min will be reserved for questions, but feel free to ask them during the class as well

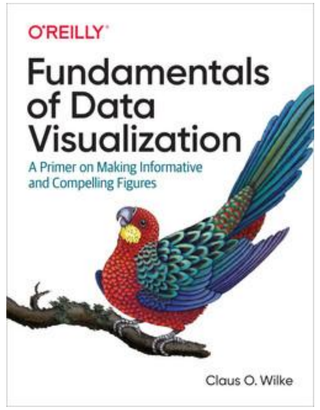
- Lectures that happen on-site will not be streamed on Zoom, live attendance is highly encouraged!
- All lectures will be recorded and put on Ilias

Material on Ilias



How to access O'Reilly e-books:
https://www.hslu.ch/en/lucerne-school-of-engineering-architecture/about-us/organization-and-topics/horw-campus/discover-campus/library/in-depth-search/

Books / Courses



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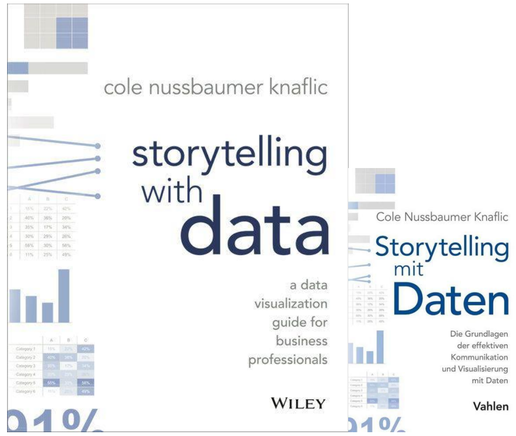
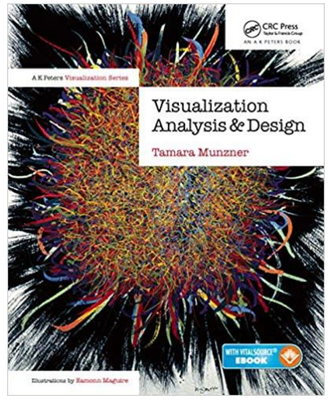
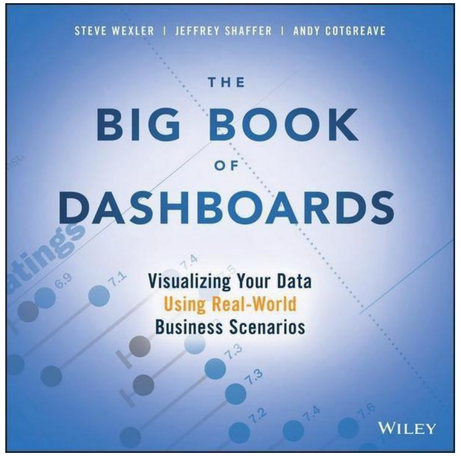


https://learning.oreilly.com/library/view/storytelling-with-data/9781119002253/



https://www.youtube.com/c/TamaraMunzner
https://learning.oreilly.com/library/view/visualization-analysis-and/9781466508910/

https://openvisualizationacademy.org/



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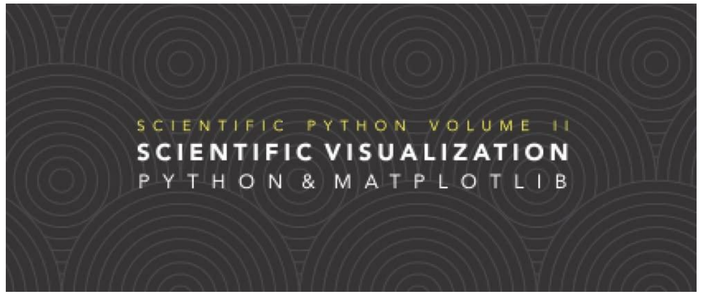
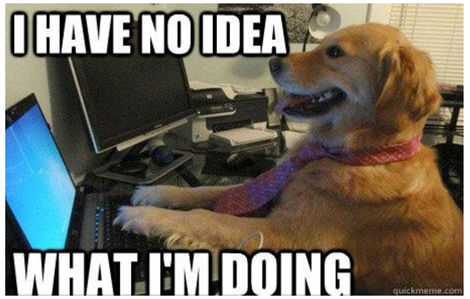


https://www.labri.fr/perso/nrougier/scientific-visualization.html

https://knowyourmeme.com/memes/i-have-no-idea-what-im-doing



Using genAI in this course

- I encourage you to try out things by yourself instead of getting the solution from AI
  - Learn the basics first, then supercharge with the AI
  - Learn to think in a particular way, not only to execute
- AI generated code for visualizations often not optimal, but sometimes good to start with
  - Imperfect training data
  - Watch out for things like hallucinated numbers
  - Learn to modify and adapt the code so that your codebase is coherent
- A lot of work in data visualization happens between humans and data, and in relation to a bigger context
  - Learn to identify and understand context, and translate it into analytics
  - Transferable and valuable soft skill
- Detailed guidelines for AI usage in your projects are in the project description

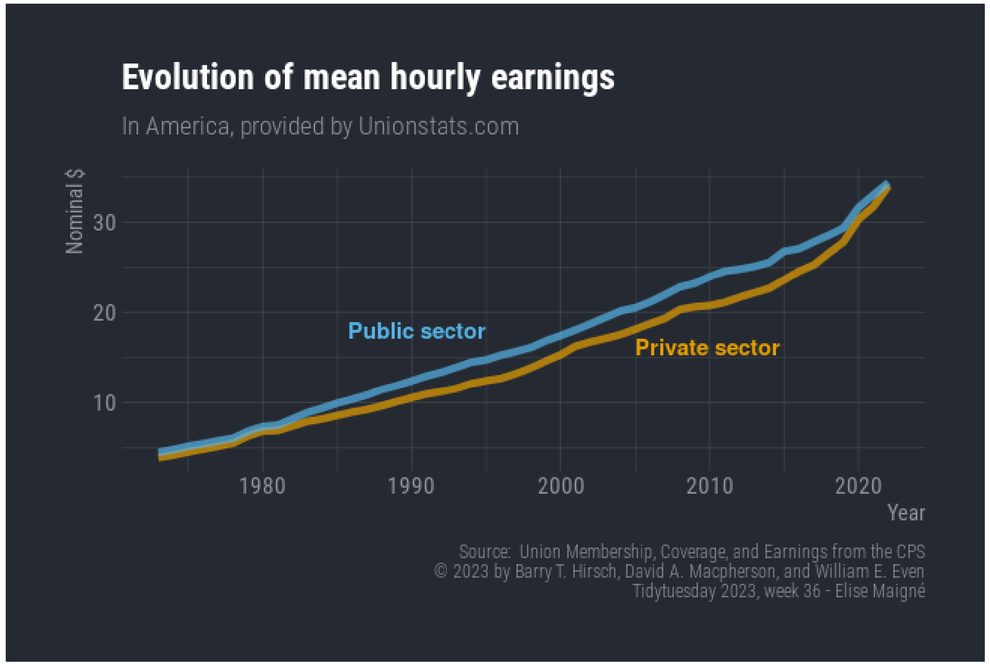
- I am interested to learn from you what worked for you and what didn’t

Grading

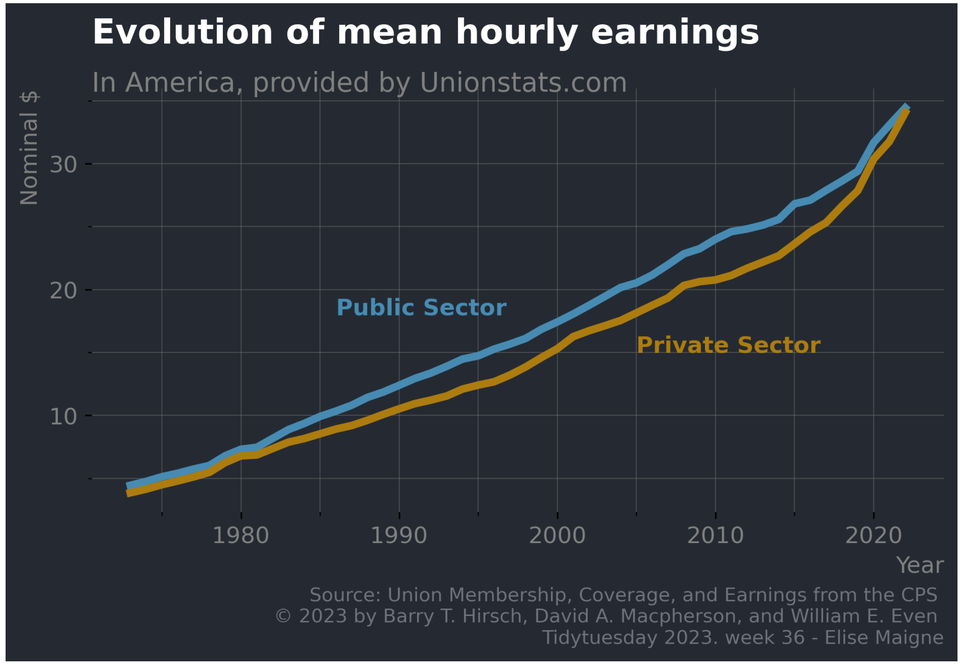
- There will be no exam
- 10% mini-project (due mid-term), 90% final project (due after the classes end but before the exam session)
- Mini-project: reproduce a visualization using Matplotlib
- Final project: a project and a ca. 4-6 pages report in English
- Course grading:
  - 10% mini-project
  - 50% final project
  - 40% final report
- Individual detailed feedback – **on request**
- Detailed info in separate document (ILIAS)

Mini-project example

original



a “good-enough” recreation



https://twitter.com/EliseMaigne/status/1699409444612764107

Final project examples

Dollar-cost averaging

Before reaching a conclusion about investing in the UBS ETF (CH) SMI, we will also consider a second investment strategy called dollar-cost averaging (DCA). This is a more sophisticated strategy than the one we used before, and it involves investing a fixed amount of money in the underlying asset at regular intervals to reduce the impact of volatility. In this case, we will start with an initial investment of CHF 10,000, and then invest an additional amount of CHF 500 every month.

As we discussed earlier, the worst time to invest in the UBS ETF (CH) SMI was in 2007 at its all-time high. Using the previous investment strategy, it would have taken more than 6 years to be in profit. However, what would have happened if we had used the DCA strategy instead?

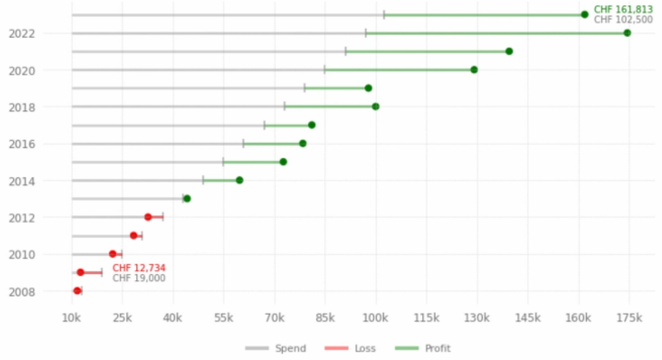


Figure 7: Investment performance from 06/2007 to 12/2012 using DCA

The grey lines in the chart indicate the total amount spent up to that point in time using the DCA strategy, as we are continuously investing more funds. As we can see, the investment would have also been underwater initially, which is not surprising given that we bought at the peak of the market

Age Distribution

Let's start by taking a look at the age distribution of the people registered at the Swiss Badminton association. The chart below makes a difference between two kinds of badminton players. Firstly, there are the **licensed** players, who can participate in as many ranked tournaments as they want, play for their club against other clubs, and basically are the active players. The second category are the **unlicensed**, players, who can't participate in ranked tournaments, or only in a limited amount, and thus are not really the active players of the community.

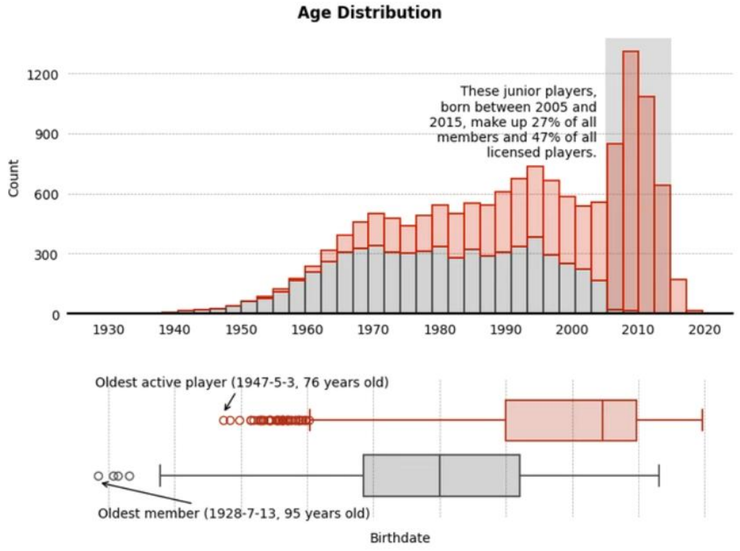
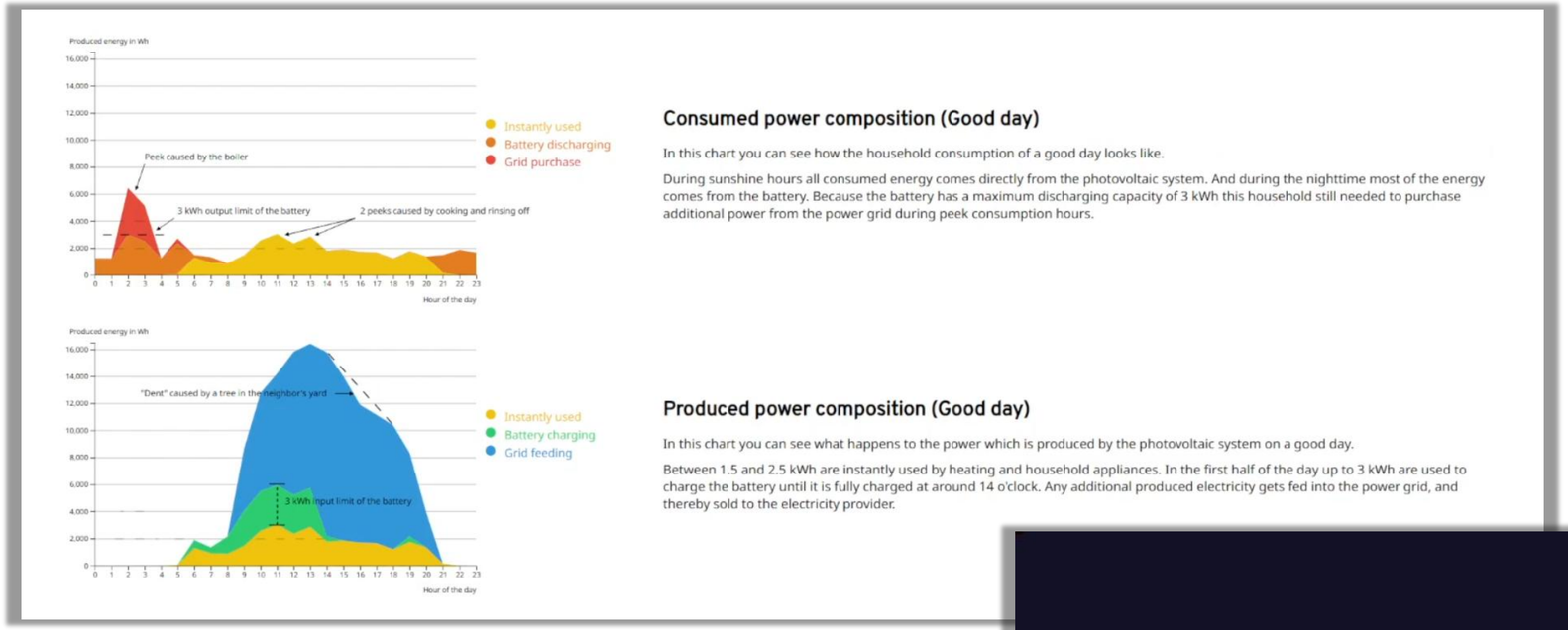
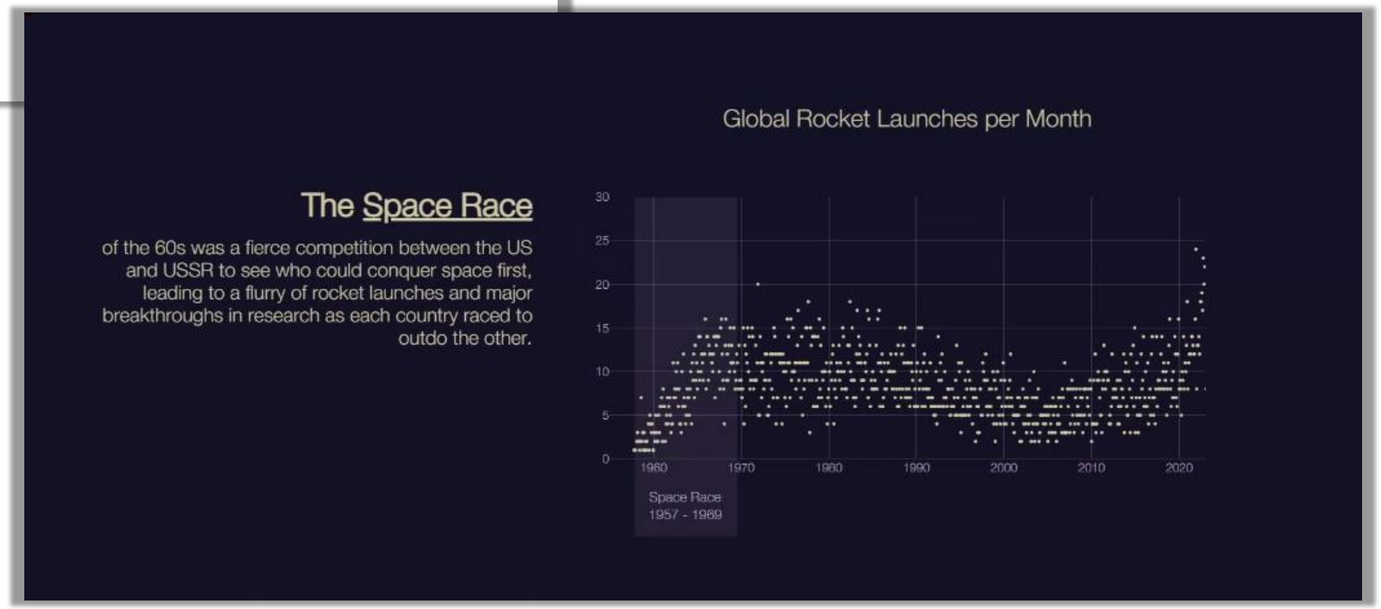


Figure 1: Age distribution of the Swiss Badminton members.

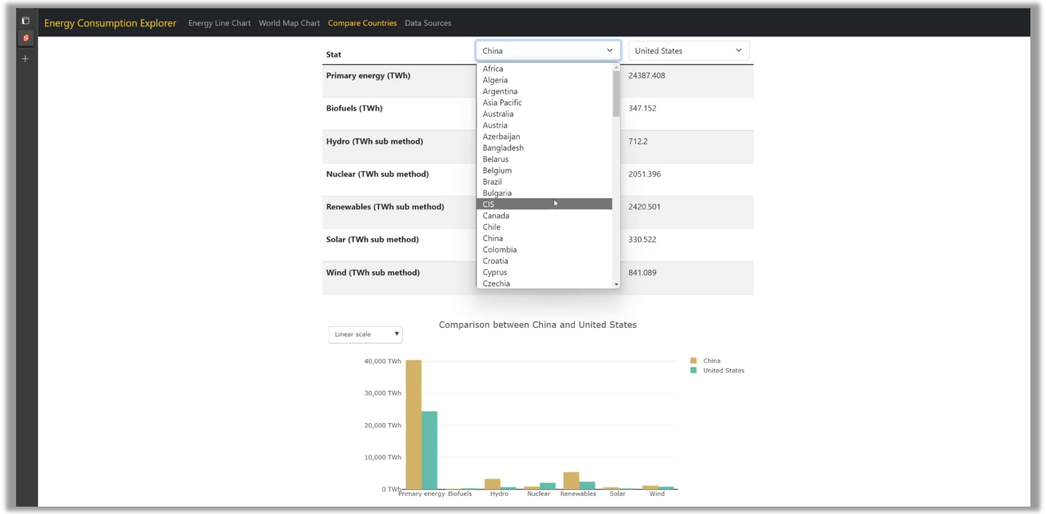
The very first characteristic that may catch your eye is the spike in members born between 2005 and 2015. They alone make up 47% of the **licensed** badminton players. With the median birthdate being mid 2004, the **licensed** badminton community is a pretty young one. This actually is the complete opposite of what I expected from personal experience at ranked tournaments, where most of my opponents were older than me, yet I am already older than the median age. A very likely explanation to my personal observation, is that these junior players mainly participate in Ranked Tournaments for Juniors (RTJ) and not in the standard

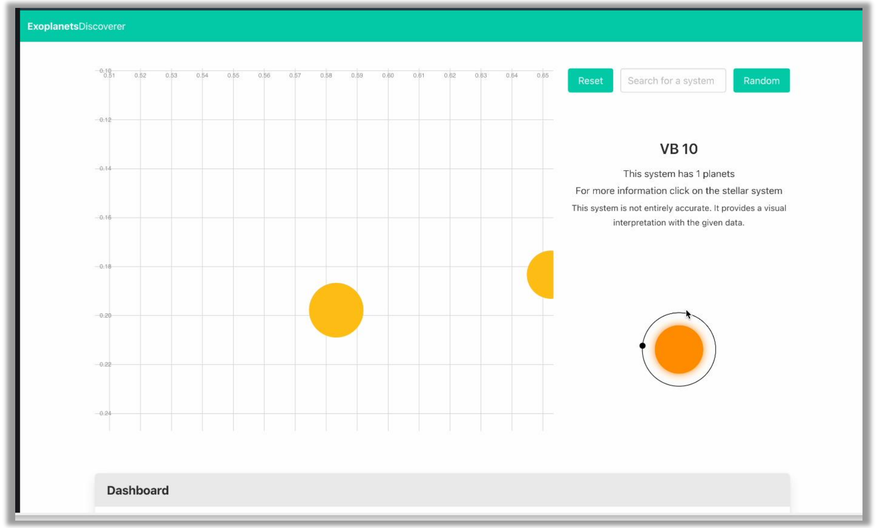
Final project examples

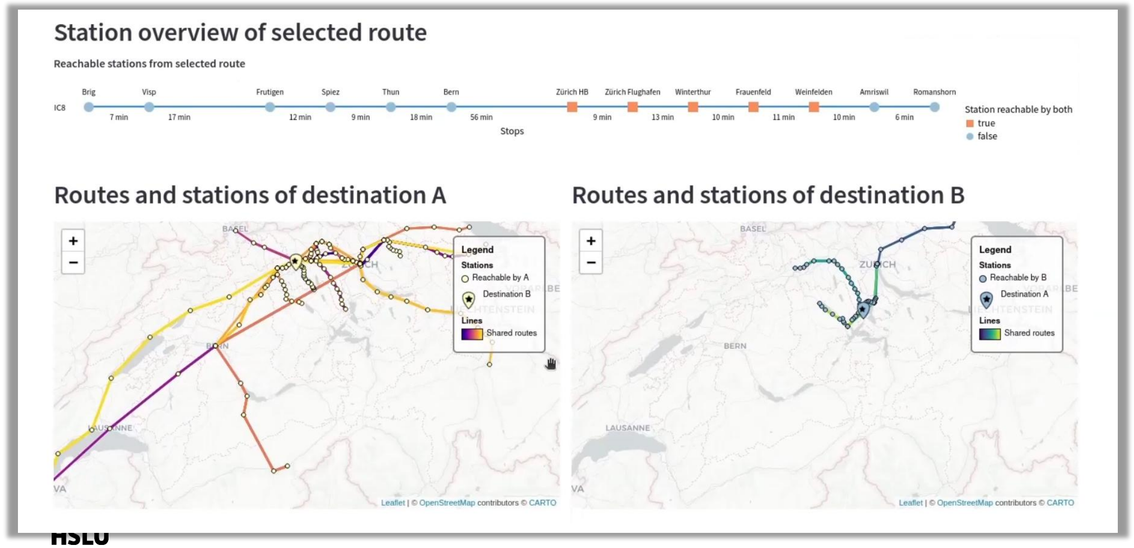


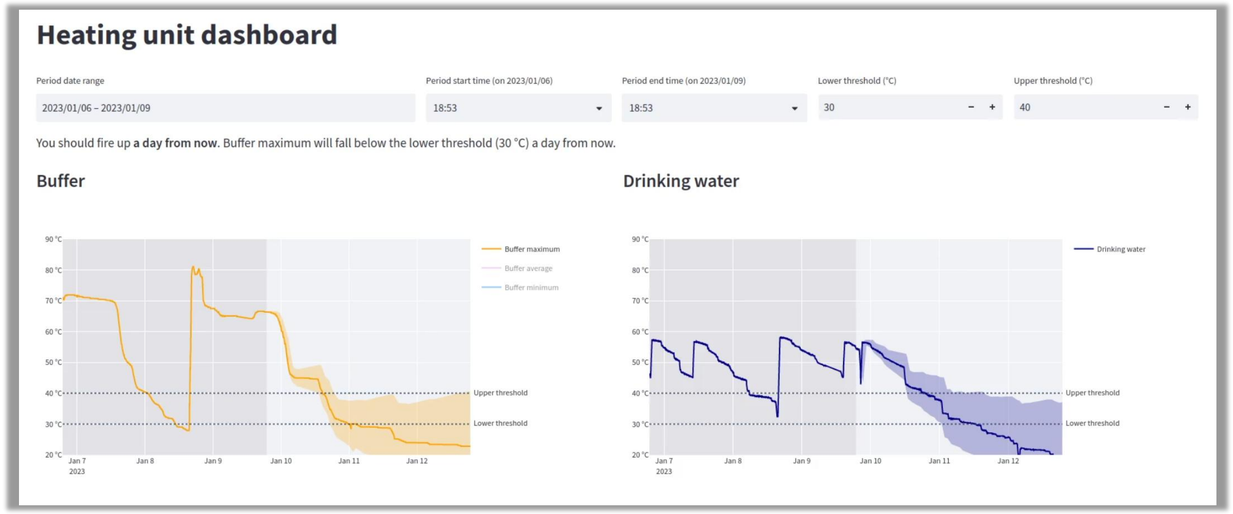


Final project examples









Additional announcements

- Every now and then I may post an extra announcement on the forum about job opportunities, dataviz competitions etc. – keep eye on them!
- Consider polishing up your final project and submitting it to the IIB Awards next summer https://www.informationisbeautifulawards.com/